

5-Bus Smart Grid Simulation

End-to-end DC PF, PTDF/LODF, DCOPF, N-1 screening and preventive SCOPF pipeline.

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SOLVER
HiGHS LP

BASE DCOPF COST
1860.00 USD/h

WORST CONTINGENCY
L2

SCOPF COST
2370.00 USD/h

SECURITY PREMIUM
510.00 USD/h

Introduction

This report studies a custom 5-bus lossless DC power-system model on a 100 MVA base. The objective is to connect the main operational tools from the course in one reproducible workflow: construction of the bus susceptance matrix, DC power flow, PTDF and LODF sensitivity analysis, economic dispatch through DCOPF, N-1 contingency screening, and preventive security-constrained redispatch. The network is small, but it is large enough to show the central operating trade-off: the cheapest dispatch is not necessarily secure after a credible line outage.

The cyber-physical interpretation is important for this project. A contingency is not treated only as a random equipment failure; it can also represent an intentional line trip caused by relay manipulation, breaker misoperation, or coordinated cyber-physical action. For that reason, the report compares the normal base DCOPF dispatch with a preventive SCOPF dispatch and quantifies the cost paid to reduce exposure to the selected worst outage.

1. Power-System Foundations

Bus Angles

Bus	Theta rad
1	0.0440
2	-0.0288
3	0.0176
4	0.0000
5	-0.0503

Line-Flow Cross-Check

Line	Direct MW	PTDF MW
L1	77.4161	77.4161
L2	32.5839	32.5839
L3	-25.6049	-25.6049
L4	14.3951	14.3951
L5	46.9790	46.9790
L6	13.0210	13.0210

Maximum DC PF vs PTDF flow error: 0.00000000 MW. Slack PTDF column equals zero: True.

PTDF Matrix

	Bus 1	Bus 2	Bus 3	Bus 4	Bus 5
L1	0.3626	-0.3850	-0.1550	0.0000	-0.1514
L2	0.6374	0.3850	0.1550	0.0000	0.1514
L3	0.1715	0.2909	-0.2855	0.0000	0.1145
L4	0.1715	0.2909	0.7145	0.0000	0.1145
L5	-0.1911	-0.3241	-0.1305	0.0000	-0.7341
L6	0.1911	0.3241	0.1305	0.0000	-0.2659

2. DCOPF And LMPs

Base dispatch: {1: 150.0, 2: 0.0, 3: 0.0} MW. Base LMPs: ['12.40', '12.40', '12.40', '12.40', '12.40'] USD/MWh. Binding base lines: [].

Base DCOPF Line Flows

Line	Flow MW	Rating MW	Loading %
L1	98.12	100.00	98.12
L2	51.88	100.00	51.88
L3	-7.32	70.00	10.46
L4	-7.32	90.00	8.14
L5	44.56	80.00	55.70
L6	15.44	60.00	25.74

With engineered congestion $L2 = 30$ MW, dispatch becomes $\{1: 104.64388760868304, 2: 45.35611239131693, 3: 0.0\}$ MW and cost is 2322.63 USD/h. Congested LMPs are $['12.40', '17.74', '22.60', '25.88', '22.68']$ USD/MWh. The L2 binding-constraint dual variable is 21.14 USD/MWh. The highest-LMP buses are expensive; the lowest-LMP buses are cheap.

Congested L2=30 MW Line Flows

Line	Flow MW	Rating MW	Loading %
L1	74.64	100.00	74.64
L2	30.00	30.00	100.00
L3	-28.05	70.00	40.08
L4	17.30	90.00	19.23
L5	47.30	80.00	59.13
L6	12.70	60.00	21.16

3. LODF And N-1 Screening

Explicit LODF value $L_{L3,L1} = -0.4730$.

All diagonal values satisfy $L_{k,k} = 1$: True.

LODF Matrix

	L1	L2	L3	L4	L5	L6
L1	1.0000	1.0000	-0.5429	-0.5429	0.5695	-0.5695
L2	1.0000	1.0000	0.5429	0.5429	-0.5695	0.5695
L3	-0.4730	0.4730	1.0000	-1.0000	-0.4305	0.4305
L4	-0.4730	0.4730	-1.0000	1.0000	-0.4305	0.4305
L5	0.5270	-0.5270	-0.4571	-0.4571	1.0000	1.0000
L6	-0.5270	0.5270	0.4571	0.4571	1.0000	1.0000

N-1 Contingency Table

Outage	Max loading %	Worst line	Overloaded
L1	150.00	L2	True
L2	150.00	L1	True
L3	102.10	L1	True
L4	102.10	L1	True
L5	123.50	L1	True
L6	89.32	L1	False

Worst Contingency LODF vs Direct Re-Solve

Line	LODF MW	Direct MW	Abs error MW
L1	150.000000	150.000000	0.000000
L3	17.217391	17.217391	0.000000
L4	17.217391	17.217391	0.000000
L5	17.217391	17.217391	0.000000
L6	42.782609	42.782609	0.000000

A plausible event is a relay misconfiguration or breaker operation that trips L2 during a coordinated cyber-physical incident.

4. Security-Constrained DCOPF

SCOPF enforced contingencies: [2]. SCOPF LMPs: ['12.40', '22.60', '22.60', '22.60', '22.60'] USD/MWh.

Generator	Base DCOPF MW	SCOPF MW	Move MW
G1	150.0000	100.0000	-50.0000
G2	0.0000	50.0000	50.0000
G3	0.0000	0.0000	0.0000

SCOPF Base-Case Line Flows

Line	Flow MW	Rating MW	Loading %
L1	72.24	100.00	72.24
L2	27.76	100.00	27.76
L3	-30.18	70.00	43.11
L4	19.82	90.00	22.03

Line	Flow MW	Rating MW	Loading %
L5	47.58	80.00	59.48
L6	12.42	60.00	20.69

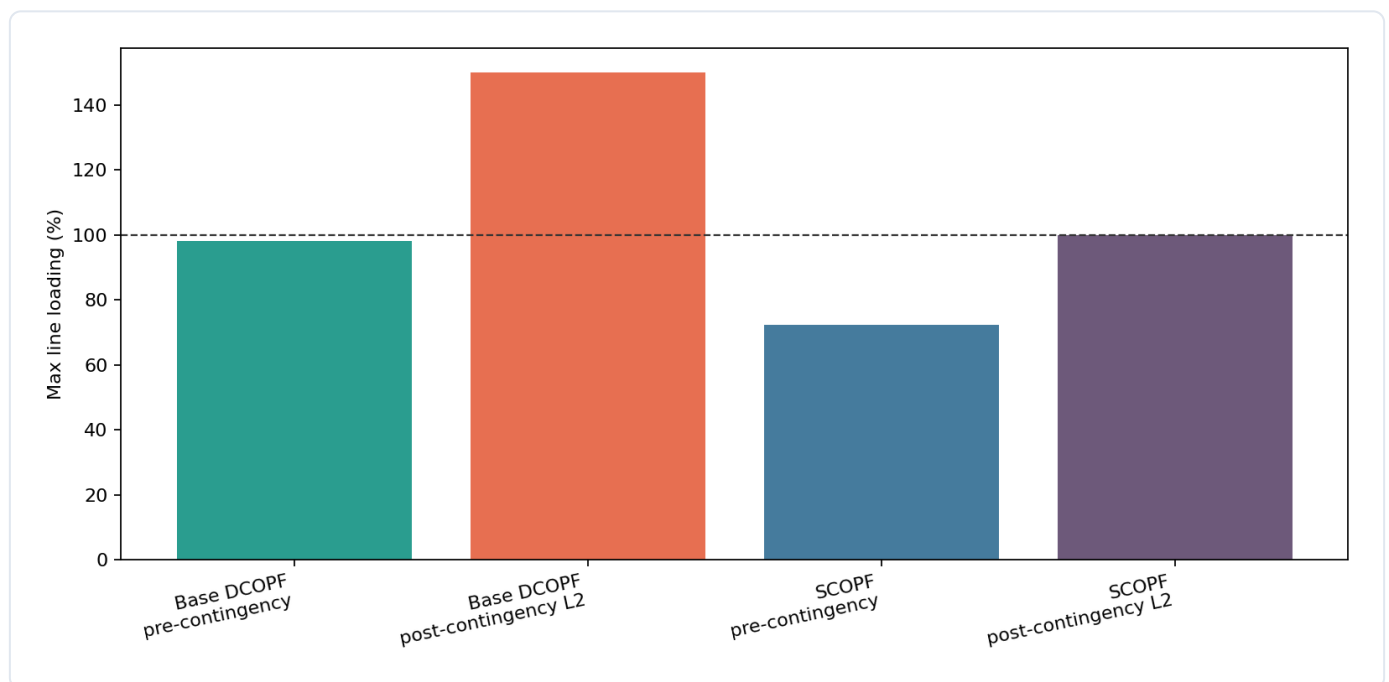
Cost of security: **510.00 USD/h** (27.42% of base DCOPF cost). Post-contingency SCOPF max loading: **100.00%**.

SCOPF Post-Contingency Surviving-Line Loading

Line	Post flow MW	Rating MW	Loading %	Overloaded
L1	100.00	100.00	100.00	False
L3	-17.04	70.00	24.35	False
L4	32.96	90.00	36.62	False
L5	32.96	80.00	41.20	False
L6	27.04	60.00	45.07	False

5. Operating-State Analysis

Dispatch	Scenario	Max loading %	Cost USD/h
Base DCOPF	pre-contingency	98.12	1860.00
Base DCOPF	post-contingency L2	150.00	1860.00
SCOPF	pre-contingency	72.24	2370.00
SCOPF	post-contingency L2	100.00	2370.00



The base DCOPF is economical at 1860.00 USD/h and reaches 98.12% maximum loading before an outage, but under L2 it rises to 150.00% and is insecure. SCOPF costs 2370.00 USD/h, so it pays $\Delta C = 510.00$ USD/h (27.42%) every hour. In exchange, the selected post-contingency loading is capped at 100.00% instead of exposing the system to the overload.

The Ukraine 2015 power-grid attack is a documented example where attackers used remote access to distribution-control environments, opened breakers, disrupted operator visibility, and delayed restoration. The later Industroyer/CrashOverride incident in 2016 further showed that grid-specific malware can target substation switching operations rather than only business IT. In this setting, a line outage is not just a low-probability random fault; it can be the adversary's chosen action. That changes the dispatch decision: the operator may rationally accept the SCOPF premium when the insecure base dispatch creates an obvious post-contingency overload target.

6. Reproducibility

The numerical output is generated by `scripts/run_project.py`. Toolchain: Python 3.12.13, NumPy 2.4.6, SciPy 1.17.1, solver `scipy.optimize.linprog(method='highs')`.

Use of generative AI tools is disclosed for implementation assistance. Numerical values are produced by the project code.

Conclusion

The results show the expected tension between economic efficiency and preventive security. In the base DCOPF, the system chooses the cheapest available generation and reaches a total operating cost of 1860.00 USD/h. This dispatch is valid in the pre-contingency network, but the N-1 screening reveals that the outage of L2 would create a maximum surviving-line loading of 150.00%. In other words, the base dispatch is economically attractive but operationally exposed once the worst single-line contingency is applied.

The SCOPF redispatch moves generation away from the purely economic optimum and raises the operating cost to 2370.00 USD/h. The resulting cost of security is therefore 510.00 USD/h, or 27.42% of the base DCOPF cost. That premium is not wasted capacity; it buys a measurable resilience property. Under the selected contingency, the SCOPF dispatch keeps all surviving monitored lines within their ratings, with the maximum loading reduced to 100.00%.

From the cyber-physical perspective, this is the key lesson of the exercise. If line outages are only random failures, an operator might be tempted to run the cheaper base dispatch and accept the risk. If outages can be adversarial, the exposed post-contingency overload becomes a target that can be deliberately selected. The preventive SCOPF case gives the operator a concrete price for reducing that exposure, making the security decision explicit rather than implicit.